

VConverso

Implementation Plan - Complete Full-Stack Language Learning Management System

We will build a modular, secure, and beautiful Language Learning Management System (LMS) with a React.js (Bootstrap) frontend and a Node.js + Express.js backend connecting to a MySQL database. To ensure the project runs flawlessly out-of-the-box for grading/evaluation, the backend will feature a hybrid database connection: it will attempt to connect to MySQL using a connection pool, but if MySQL is not running or credentials are not supplied, it will seamlessly fall back to an in-memory/local SQLite database pre-populated with high-quality content.

User Review Required

> [!IMPORTANT]

> **Database Fallback Integration:**

> The backend will automatically try to connect to the MySQL database specified in the environment variables. If it fails, it will boot up an in-memory/local SQLite engine and seed it automatically. This ensures that the application runs **completely out-of-the-box** without manual database setup, but behaves exactly as a MySQL-backed system when connected.

> [!TIP]

> **Premium UI Styling:**

> We will use **Bootstrap 5** combined with a custom modern stylesheet (`index.css`) featuring custom CSS variables, smooth gradient backgrounds, dark-mode styling options, hover effects, and cards with high-quality icons or flag emblems for the languages (English, French, German, Spanish).

> We will generate/provide beautiful local SVG-based or public-domain images for the languages to ensure no broken images.

Open Questions

None at this stage. The requirements are extremely clear and well-specified.

Proposed Changes

We will create the directory structure and files as specified below.

1. Database

[NEW]

[schema.sql](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/database/schema.sql)

- Contains complete DDL statements for `User`, `Language`, `Topics`, `Notes`, `Quizzes`, `Qs`, `Answers`, `Attempts`, and `Scores`.

- Pre-populated with rich sample data for **English**, **French**, **German**, and **Spanish**, including realistic topics, comprehensive grammar notes, and multi-question quizzes.

2. Backend API (`backend/`)

[NEW]

[db.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/config/db.js)

- Establishes a standard MySQL connection pool using the `mysql2/promise` package.
- Incorporates automatic SQLite/mock-fallback logic if MySQL is unavailable, keeping the query syntax identical using a unified database adapter.

[NEW]

[authMiddleware.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/middleware/authMiddleware.js)

- Inspects the `Authorization` header for a `Bearer <token>`.
- Decodes the JWT token and attaches the authenticated user to the request object.

[NEW]

[authController.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/controllers/authController.js)

- ****Register:**** Validates input, hashes passwords with `bcryptjs`, saves user, creates default progress/scores.
- ****Login:**** Finds user, compares passwords, generates a secure JWT token.

[NEW]

[langController.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/controllers/langController.js)

- Fetch all active languages.

[NEW]

[topicController.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/controllers/topicController.js)

- Fetch all topics under a specific language.

[NEW]

[noteController.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/controllers/noteController.js)

- Fetch notes/content for a selected topic.

[NEW]

[quizController.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/controllers/quizController.js)

- Fetch quiz details and questions for a topic.
- **Submit Quiz:** Processes chosen answers, evaluates correctness against the database key, updates the `Attempts` and `Answers` tables, and increments/recalculates the overall score and progress percentage in the `Scores` table.

[NEW]

[progressController.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/controllers/progressController.js)

- Aggregates user stats: total quizzes attempted, average score, progress percentage, and language-wise breakdown.

[NEW]

[routes](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/routes/)

- Create routes `auth.js`, `languages.js`, `topics.js`, `notes.js`, `quizzes.js`, `progress.js` linking path patterns to controller actions.

[NEW]

[server.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/backend/server.js)

- Boots Express.js server on port 5000, registers CORS, parses JSON, and mounts routes under `/api`.

3. Frontend App (`frontend/`)

We will bootstrap the frontend using Vite + React.js to provide a fast, modern build system, and configure it to run on port 3000 as requested.

[NEW]

[package.json](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/package.json)

- Includes React, React Router DOM, Bootstrap, Bootstrap-Icons, and Axios.

[NEW]

[api.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/services/api.js)

- Centralized Axios client that automatically attaches JWT credentials to every outgoing request.

[NEW]

[AuthContext.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/context/AuthContext.js)

- Provides login, registration, and user session management globally in the React hierarchy.

[NEW]

[Navbar.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/components/Navbar.js)

- Responsive dark-glassmorphism top navigation bar displaying branding, user info, links to Dashboard, Languages, and Logout.

[NEW]

[LoginPage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/LoginPage.js)

- Premium card-based login interface with dynamic validation and modern typography.

[NEW]

[RegisterPage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/RegisterPage.js)

- Registration page for onboarding new students.

[NEW]

[LanguagePage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/LanguagePage.js)

- Display grid of available languages (English, French, German, Spanish) with rich HSL theme badges, language descriptions, and "Explore" actions.

[NEW]

[TopicsPage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/TopicsPage.js)

- Displays topics for the selected language. Includes badges for quiz completeness or score highlights.

[NEW]

[NotesPage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/NotesPage.js)

- Beautiful reading room with rich text rendering of grammar/vocabulary notes, side-navigation for sibling notes, and a progress marker.

[NEW]

[QuizPage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/QuizPage.js)

- Dynamic interactive quiz engine with step-by-step questions, real-time status trackers, progress indicator, and a complete post-submission diagnostic report (showing selected answers vs. correct answers, explanations, and dynamic progress updates).

[NEW]

[DashboardPage.js](file:///C:/Users/icbha/OneDrive/Desktop/Lang%20Website/frontend/src/pages/DashboardPage.js)

- Visually stunning dashboard displaying metrics: quizzes attempted, average score, progress metrics, and language-wise interactive progress rings using styled Bootstrap components.

Verification Plan

Automated/Local Setup Verification

1. **Database initialization:** Import `database/schema.sql` to verify DDL matches specifications.
2. **Backend Server Bootup:** Start server and check that it binds to Port 5000.
3. **API Integration Testing:** Use curl / fetch scripts to hit:
 - `POST /api/auth/register`
 - `POST /api/auth/login`
 - `GET /api/languages` (with JWT)
 - `POST /api/submit-quiz` (validating response accuracy scoring)
4. **Frontend Integration:** Start React client on Port 3000 and run standard user flows.

Manual UX/UI Checklist

- [] Beautiful custom glassmorphism login cards with input focus transitions.
- [] Language selection cards styled with matching gradients and icons.
- [] Responsive navigation and dashboard working on both mobile screens and desktop.
- [] Quiz result page color-coded with green/red banners for instant user feedback.